
Brief – Innovation Semester

Framework: *Game for Change* × IoT × Art-Based Climate Awareness

1. Conceptual Foundation

This innovation semester will be structured around the **Games for Change framework**.

Games for Change promotes the idea that games and playful systems can:

- Raise awareness about social and environmental challenges
- Influence behaviors
- Stimulate civic engagement
- Foster empathy and systems thinking

The semester challenge translates this philosophy into a **physical, interactive, IoT-enabled art installation** addressing global sustainability challenges.

Learning Outcomes (CTI-Compatible Formulation)

At the end of the semester, students will be able to:

- Design and prototype an IoT-based interactive system addressing a UN Sustainable Development Goal.
- Integrate scientific modeling, embedded systems, and data interpretation into a functional prototype.
- Apply systems thinking to sustainability challenges.
- Develop innovative solutions in uncertain and multidisciplinary environments.
- Communicate complex socio-technical systems to non-specialist audiences.

- Manage a full innovation cycle from concept to exhibition-ready prototype.
- Integrate ethical and environmental responsibility into engineering design.
- Demonstrate collaborative engineering practices in multidisciplinary teams.

2. Core Challenge

Students must design and prototype an **interactive art-based game system** that:

1. Is rooted in one or several of the **United Nations Sustainable Development Goals (SDGs)**
2. Integrates **IoT (Internet of Things)** components
3. Includes a **physical interactive element**
4. Uses **game mechanics or gamification principles**
5. Functions as both:
 - A playable or interactive experience
 - A contemporary art installation suitable for exhibition

The final output must be exhibited as:

- A **Game Show experience**
- A **Modern Art-style exhibition installation**

3. Thematic Anchoring: SDGs

Projects must explicitly reference and integrate one or more SDGs (the more systemic integration, the stronger the proposal).

Examples:

- SDG 13 – Climate Action
- SDG 12 – Responsible Consumption
- SDG 11 – Sustainable Cities
- SDG 6 – Clean Water
- SDG 7 – Clean Energy

Students must:

- Identify the SDG(s)

- Define the specific issue
- Translate it into a system-level problem
- Express it through interactive mechanics

4. Design Constraints

4.1 Game for Change Logic

The project must:

- Create awareness
- Trigger reflection
- Potentially influence behavior
- Engage through play

The “game” can be:

- Competitive
- Cooperative
- Immersive
- Reflective
- Experiential

Gamification elements may include:

- Points
- Feedback loops
- Narrative
- Rewards
- Role-playing
- System simulation
- Real-time data interaction

4.2 IoT Integration (Mandatory)

Each project must include **a physical IoT device** that:

- Senses something (temperature, CO₂, motion, light, sound, pollution, water usage, human interaction, etc.)
- Interacts physically (buttons, levers, actuators, vibration, motors, LEDs, screens, projection)
- Connects data (local or cloud-based)
- Materializes invisible phenomena

The IoT device must serve the concept — not be decorative.

Examples:

- A sculpture reacting to real CO₂ levels
- A city model that lights up based on energy consumption data
- A wearable artifact that changes color with environmental metrics
- A tactile interface simulating resource depletion

4.3 Artistic Dimension (Mandatory)

The system must be conceived as an **art installation**, not just a technical prototype.

It must:

- Have aesthetic coherence
- Convey meaning symbolically
- Engage at sensory levels

Students may use:

- Visual expression
- Sound
- Touch
- Smell
- Taste (if feasible)
- Spatial design
- Materiality

It should be exhibit-ready in a contemporary art context.

5. Expected Format of Final Deliverable

Exhibition Format

At the end of the semester, projects will be presented in two formats:

1. Game Show Format

- Public interaction
- Demonstration of mechanics
- Player participation
- Live feedback

2. Museum-Style Installation

- Autonomous installation
- Curated display
- Artistic statement
- Concept explanation panel

The installation must:

- Be stable and functional
- Be understandable by non-experts
- Communicate its SDG relevance

6. Interdisciplinary Integration

The semester integrates multiple courses:

- IoT engineering
- Interaction design
- Systems thinking
- Sustainability studies
- Art & design
- Game design
- Entrepreneurship / innovation

Students must demonstrate:

- Technical robustness
- Conceptual coherence
- Narrative clarity
- Societal relevance

7. Innovation Expectations

Projects should aim to:

- Combine physical computing with meaningful storytelling
- Translate complex sustainability issues into embodied experience
- Make invisible environmental data tangible
- Create emotional and cognitive engagement

We expect:

- Functional prototype
- Tested interaction loop
- Clear user experience design
- Documented system architecture

8. Evaluation Axes (to be formalized academically)

Suggested evaluation pillars:

1. SDG integration depth
2. Game mechanics quality
3. IoT implementation robustness
4. Artistic coherence
5. Interaction design
6. Systems thinking
7. Public engagement capacity
8. Technical documentation

9. Philosophical Positioning

This semester positions students not only as engineers but as:

- System designers
- Cultural actors
- Technology mediators
- Sustainability interpreters

The objective is not merely to build a device — but to build a **meaningful interactive experience that changes perception**.

Team Formation Process for the Innovation Course

At the start of the course — especially with large cohorts of around 100 students — we bring everyone together in one room.

First, we give a brief recap of the overall course theme or topic.

Then, each person who wants to lead a team gets one minute to pitch their idea: what they want to work on, what the project is about, and why others should join them.

After each pitch, the team leader stands at a designated spot in the room.

Once all pitches are done, we invite all other students to walk around and physically gather behind the leader they want to work with.

However, teams must meet three requirements:

- Gender diversity: Each team of five must include at least two genders.
- Skill diversity: Teams should bring together different skills and backgrounds to ensure they can successfully carry out the project.
- School diversity: Each team should have members from at least 3 universities.

Students are encouraged to check who is gathering where and adjust to make sure their team is diverse enough.

Once the groups are formed, each team must post in the course WeChat group:

- The list of the five team members
- The name of their group

This creates spontaneous, interest-based teams while ensuring diversity and making the process interactive and visible for everyone.